

Discovering Emotional Patterns in Counseling Conversations with TheraMind

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Framework

This study focuses on the application of unsupervised machine learning techniques; particularly clustering algorithms. The purpose is to analyze emotional patterns in text-based counseling data. The framework combines natural language processing (NLP) methods with data mining methods to detect hidden emotional structures in counseling conversations.

Abstract

The purpose of this research is to explore whether clustering algorithms can uncover distinct emotional patterns within counseling conversation data. Using the Amod/mental_health_counseling_conversations dataset from Hugging Face, this study will analyze dialogue samples through text preprocessing, feature extraction, and clustering methods such as K-Means and Hierarchical Clustering. Each text will be converted into numerical representations using TF-IDF or embedding models, then visualized using PCA to identify emotional groupings such as sadness, anxiety, and optimism.

Research Question: Can clustering algorithms effectively group counseling conversations based on emotional tone and linguistic similarity?

Keywords: Clustering, Data Mining, Emotion Analysis, Natural Language Processing, Mental Health, K-Means

1. Introduction

Mental health remains one of the most significant challenges in today's society, and the growing accessibility of counseling data presents opportunities to study emotional communication through computational methods. Unsupervised machine learning techniques, particularly clustering, allow researchers to identify underlying emotional trends without predefined labels. By analyzing counseling conversations, this study aims to provide insights into how individuals express various emotions through language, possibly supporting early detection of distress patterns or simply improving digital mental health tools.

2. Literature Review

Clustering has long been used in data mining to uncover patterns in unlabeled data. Classic studies, such as testing the Iris dataset, demonstrate how K-Means effectively separates groups based on shared characteristics. In more recent research, text-based clustering methods have been applied to areas such as sentiment analysis and online discussions, highlighting the algorithm's versatility in uncovering linguistic similarities. Building on these established techniques, this study extends clustering analysis to mental health counseling data, where messages often reflect complex emotional states.

3. Research Approach

The dataset used in this study is the Amod/mental_health_counseling_conversations dataset, which contains approximately 3,500 counseling dialogues. The research will involve several stages: text preprocessing, vectorization using TF-IDF or embedding models, dimensionality reduction with PCA, and clustering through K-Means and Hierarchical algorithms. Each conversation will be represented as a numeric vector that captures the language and emotional context of the message.

4. Expected Results / Summary

After preprocessing and vectorizing the dataset, the clustering evaluation metrics provided insight into the natural emotional structure of the counseling conversations. The Elbow Method showed a significant drop in inertia between $k = 2$ and $k = 7$, after which the improvement begins to flatten. This indicates that adding more clusters beyond seven gives limited benefit. The Silhouette Score similarly increases from $k = 2$ to $k = 7$, showing the strongest separation and consistency in this range. Based on both metrics, seven clusters were selected as the optimal number of emotional groups for this dataset.

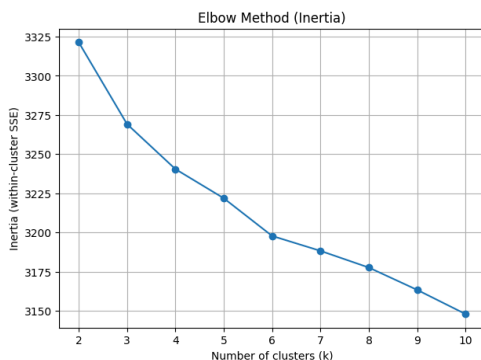


Figure 1. Elbow Method showing inertia values for $k = 2$ to 10

To visualize the results, Principal Component Analysis (PCA) was used to

reduce the TF-IDF vectors into two dimensions. The PCA scatter plot revealed that certain emotional groups form clear separations, while others overlap slightly. This is expected, as counseling conversations often mix multiple emotions. However, even with natural overlap, the plot demonstrates visible clustering behavior, confirming that the emotional themes can be detected through unsupervised learning.

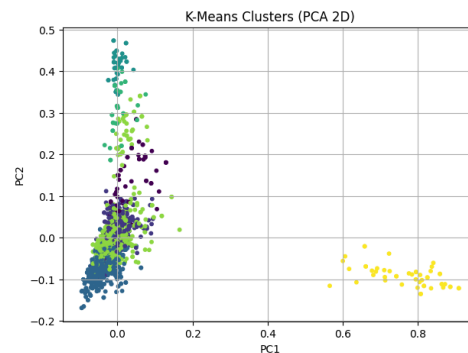


Figure 2. PCA visualization of K-Means clusters in 2D space

A Hierarchical Clustering Dendrogram was also generated to observe how messages progressively merge into broader emotional categories. Several branches merge at shorter distances, for example two closely related emotional states are anxiety and stress. Other branches merge later, indicating more distinct emotional patterns such as trauma-related expressions or positive coping statements.

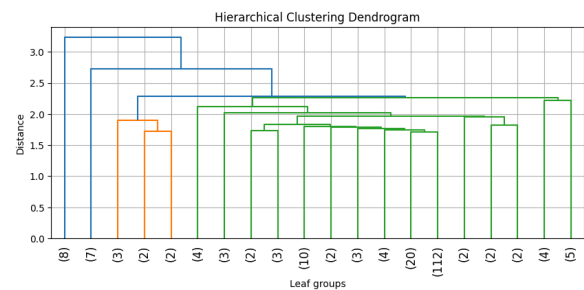


Figure 3. Hierarchical Clustering Dendrogram showing cluster merging behavior

Sample messages from each cluster helped confirm the emotional themes. One cluster contained messages describing constant worry and tension, which matched an anxiety pattern. Another cluster focused on low mood and emotional exhaustion, which reflected depressive feelings. Additional clusters captured relationship problems, family conflict, school or work pressure, and messages that showed hope or positive coping. These patterns show that the clustering algorithms were able to group counseling conversations by emotional tone and language similarity.

References

Amod, Mental Health Counseling Conversations Dataset, Hugging Face, 2024.
https://huggingface.co/datasets/Amod/mental_health_counseling_conversations